

From Laboratory to Field: A Leakage-Controlled Cross-Domain Evaluation Protocol for Plant Disease Recognition

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Abstract. Plant disease classifiers trained on laboratory imagery report accuracies above 0.99, yet field deployment disappoints. The size of that gap is widely asserted and rarely measured under conditions that make the measurement trustworthy. We report a cross-domain evaluation in which the two obvious confounds are closed rather than assumed away. First, train/test contamination: an exhaustive perceptual-hash search over every distinct-hash pair across the two corpora, followed by human adjudication of all 16 near candidates, establishes 0 exact overlaps between the laboratory training corpus and the field evaluation corpus. Second, label-space construction: the shared 21-class space is assembled only from a deterministic normaliser applied identically to both label sets and from a previously recorded human policy decision, so no new semantic judgement is introduced to make the comparison look better or worse. Three ImageNet-pretrained backbones are trained under one protocol on the 43 596-image PlantVillage training split and evaluated on 1 951 field images across 3 seeds, with paired significance tests and bias-corrected accelerated bootstrap intervals, every input bound to a digest-verified dataset lock (83449f6013691855). Macro-F1 falls by 76.9–81.2 % under the shift. We find that near-saturated in-domain accuracy carries almost no information about which backbone transfers, because it does not produce an ordering stable enough to carry, and that a temperature fitted on in-domain validation data, standard practice for calibration, makes out-of-domain calibration substantially worse rather than better.

Keywords: Plant disease recognition · Cross-domain evaluation · Leakage audit · Model calibration · Reproducibility

1 Introduction

Image-based plant disease recognition has an unusually clean success story and an unusually persistent complaint. The success story is PlantVillage [12]: tens of thousands of labelled leaf photographs on which convolutional networks reach accuracies above 0.99, a result reproduced so widely that the benchmark is effectively saturated. The complaint is that these accuracies do not survive contact with a real field. PlantVillage images are single detached leaves against uniform backgrounds under controlled lighting; a photograph taken by a grower contains

occlusion, soil, multiple leaves, variable illumination, and a camera held at whatever distance was convenient. The gap matters because the downstream artefact is an advisory, and a confident wrong diagnosis becomes an unnecessary spray or a lost season.

That a gap exists is not in dispute, and field-condition corpora have been built precisely to expose it [18,13,21]. What is harder to find is a measurement a sceptical reader can trust, because two confounds sit between the raw comparison and any conclusion drawn from it. The first is contamination: if any field evaluation image is a near-duplicate of a laboratory training image, the measured transfer is optimistic by an unknown amount. This is not hypothetical, since such contamination has been documented in benchmarks far more scrutinised than these [1], and most cross-corpus plant disease comparisons report no leakage audit at all. The second is the label space: the two corpora do not share a taxonomy, so any cross-corpus evaluation must construct a correspondence, and that construction is a research degree of freedom. A generous mapping pairing biologically similar classes will report a smaller gap than a conservative one, and the reader usually cannot tell which was used.

This paper closes both. Our contributions are:

1. **A leakage-controlled protocol.** Duplicate candidates are generated by exhaustive perceptual-hash comparison over every distinct-hash pair rather than by an approximate index, and every candidate is adjudicated by a human.
2. **An evidence-based shared label space** assembled from exactly two pre-existing sources, with a written reason for every excluded class, including biologically plausible near-misses we decline to assert.
3. **A controlled multi-seed comparison with paired statistics.** Three backbones spanning a $15\times$ parameter range are trained under one protocol across 3 seeds and compared with McNemar’s exact test on per-item correctness [11,3] and BCa bootstrap intervals [4], with family-wise error control [9] and every verdict recomputed independently per seed.
4. **Two findings that matter for practice.** Every pairwise verdict on the saturated in-domain benchmark changes with the seed, and the backbone that ranks last on two seeds ranks first on the third, so in-domain rank is not a stable quantity from which field behaviour could be predicted. Separately, a temperature fitted on in-domain validation data and carried across the shift degrades calibration below the uncalibrated baseline, the opposite of the standard recommendation.

Where two backbones are not statistically distinguishable we say so rather than reporting their order, and where a difference is smaller than the variation the random seed alone induces, we do not claim it. Every input is bound to a digest-verified dataset lock, every reported number is generated from run artefacts by script, and no number in this document was typed by hand.

2 Related Work

Corpora, laboratory and field. PlantVillage [12] established both the standard benchmark and the standard result, accuracies above 0.99 across dozens of crop–disease classes, on imagery whose construction from detached leaves against uniform backgrounds is stated plainly in the original work. What followed was a long sequence of papers reporting incremental gains on a benchmark with very little headroom. PlantDoc [18] was assembled specifically to test whether laboratory-trained models transfer and reported that they largely do not; FieldPlant [13] and PlantWild [21] extend the setting to larger and more varied imagery, and PlantSeg [22] adds segmentation supervision. Our work is complementary: rather than proposing a new corpus or a stronger model, we ask what conditions a cross-corpus comparison must satisfy before its number means anything.

Dataset bias, leakage, and calibration. That models absorb corpus-specific regularities rather than the intended concept predates deep learning [20] and has been quantified repeatedly since [16,6]. For plant disease the uniform background of laboratory imagery is an obvious shortcut candidate, and near-duplicate leakage is the adjacent failure that [1] showed survives in benchmarks studied for a decade. Modern networks are also systematically over-confident, with temperature scaling the standard remedy [7]; under shift, uncertainty estimates degrade further and the ranking of calibration methods changes [15]. Selective prediction [5] gives the deployment-relevant view: how accurate a model is on the subset it will act on. We treat each as something to measure, not to assume.

Positioning within this venue. AgroBench [17] shows how far general-purpose vision–language models remain from reliable agricultural judgement, and the applied trajectory this community is on runs from detection through severity estimation to decision support. Our contribution sits underneath all of it: an advisory system built on a classifier selected by laboratory accuracy inherits both the wrong architecture choice and the wrong confidence estimate, and neither is visible from the leaderboard the choice was made on.

3 Datasets and Leakage-Controlled Preparation

3.1 Corpora

PlantVillage supplies the laboratory domain: 54 305 colour images over 38 crop–disease classes, split 43 596 train / 10 709 test. The corpus is pinned to an immutable upstream revision, with the manifest reconstructed exactly from the pinned split files.

PlantDoc [18] supplies the field domain. 2 578 records were acquired at a pinned source revision. This differs from the count and class total reported in the original publication, because the distributed repository has changed since;

we therefore report the pinned revision’s own numbers and bind them to a digest rather than quoting the paper’s. Two reductions follow, both recorded with reasons: 14 images are removed by human adjudication of byte-exact duplicate groups, and 3 further images are *excluded, not relabelled*, because the diagnostic evidence for a relabel was judged insufficient and the second scientific review that would settle it has not been performed. The exclusion records no diagnosis and asserts nothing about what those images depict. The resulting *PlantDoc Core* holds 2 561 images over 28 classes, split 2 336 train / 225 test. No source image was deleted and no image byte was modified.

Class support in PlantDoc Core is severely uneven. The largest training class holds 90 times the smallest, and one arthropod-pest class has two training images and no test images at all. It cannot be learned or measured, and is retained only because removing it would alter the locked corpus. Full provenance and leakage counts are in the supplementary material.

3.2 Cross-Corpus Leakage

If a field evaluation image duplicates a laboratory training image, the cross-domain number is contaminated, so we search for such pairs exhaustively. Candidate generation uses a 64-bit perceptual hash with exact chunked brute-force Hamming evaluation over *every* distinct-hash pair, which for the acquired population is 1.37×10^8 distance evaluations. An approximate nearest-neighbour index would make the absence of a match an artefact of the index’s recall, which is exactly the property this audit must not have. Pairs at Hamming distance ≤ 6 are surfaced as candidates.

Perceptual hashing generates candidates; it never decides. All 16 surfaced near pairs were adjudicated by a human as clearly different, and the great majority are cross-class collisions on plain green foliage, the failure mode a perceptual hash is expected to have. The audit finds 0 exact overlaps and 0 unresolved pairs. No laboratory training image is a byte- or pixel-duplicate of any field evaluation image, so the cross-domain measurement in Section 6 is a genuine generalisation measurement.

Within-corpus leakage is controlled separately. PlantVillage records which physical leaf each image photographs, and several images can show one leaf. The upstream train/test split is leaf-grouped; the validation split we carve from train is *also* leaf-grouped, so no physical leaf appears on both sides of model selection. Ignoring this would let a model be selected on a near-copy of something it trained on.

3.3 Why the Two Evaluation Sets Differ in Size

Two field-corpus evaluation sets appear here and are deliberately different sizes: cross-domain evaluation uses 1 951 images, in-domain PlantDoc testing 225. What sets the size is *what the model under test has already seen*. PlantDoc’s train/test boundary is a fact about PlantDoc-trained models and carries no

information about a model trained on a different corpus, so for a PlantVillage-trained model the entire shared-class subset is legitimately held out; restricting it would discard evidence for no methodological gain and leave per-class supports far too thin. A PlantDoc-trained model has seen the training split, so only the 225 held-out images are admissible, and every in-domain PlantDoc figure here uses exactly those. Neither set contains an image the model being evaluated trained on, and the cross-domain set’s composition by PlantDoc split is recorded in every result file, so a reader preferring the more restrictive choice can see which images it would remove.

4 Experimental Methodology

4.1 The Shared Label Space

The two corpora use different taxonomies, so cross-domain evaluation requires a correspondence. Because that correspondence is a research degree of freedom, we constrain it to evidence that existed before this evaluation was designed, from exactly two sources.

Source 1: deterministic key collision. The project’s crop and disease normalisation functions are applied *unchanged and identically* to both label sets. Eleven pairs collide on the resulting canonical key. The same function runs on both sides, so no human judgement enters the pairing.

Source 2: a recorded human policy. A previously approved policy assigns every PlantDoc class of the form “*Crop* leaf” a canonical disease of *healthy*; on the PlantVillage side the token is literal in the label. Ten crops appear on both sides under this policy.

The result is 21 shared canonical classes covering 1 951 PlantDoc Core images. Everything else is excluded *with a written reason*. The excluded set deliberately includes biologically plausible near-misses that the normaliser does not align, for instance a gray-leaf-spot class against a *Cercospora* gray-leaf-spot class, and two differently-named pepper leaf-spot classes. Asserting those pairs would be a new semantic judgement made by the authors of the evaluation, in a direction that would improve the headline number. We decline, and record what a more generous mapping would change; Section 9 states the consequence.

4.2 Cross-Domain Evaluation Procedure

A laboratory-trained model predicts over its own 38-way space. To evaluate it on the shared space, source-class probabilities are summed within each canonical class and renormalised over the 21 shared classes only. Field images outside the mapping are *excluded from the evaluation set* rather than counted as errors, since a model cannot be faulted for a class the protocol never asked it about. Renormalisation discards probability mass, and how much is discarded is itself informative, so the mean retained mass before renormalisation is recorded for every run and reported in Section 6. All 1 951 shared-class Core images are used, for the reason given in Section 3.3.

4.3 Models and Training

Three ImageNet-pretrained [2] backbones span an order of magnitude in capacity: ResNet-50 [8] (23.6 M parameters), EfficientNet-B0 [19] (4.0 M), and MobileNetV3-Small [10] (1.5 M). Only the final classifier layer is replaced. Backbone choice is the sole architectural variable.

One protocol governs every run: 224×224 inputs, AdamW at learning rate 3×10^{-4} with weight decay 10^{-4} , cosine decay stepped per batch, mixed precision, batch size 64, label smoothing 0.1 held fixed across every run, at most 15 epochs with early stopping on validation macro-F1 at patience 3, and selection of the single best-validation checkpoint. Augmentation is deliberately mild: random resized crop, horizontal flip, rotation within $\pm 15^\circ$, and colour jitter with *near-zero hue* perturbation, since chlorosis and necrosis are diagnosed by colour. The label-smoothing choice matters for the calibration result (Section 7).

Two deviations are dataset-driven rather than architectural, and both are reported rather than tuned. PlantVillage validation is grouped by leaf identity, as described above. PlantDoc Core training uses inverse-frequency class weighting normalised to mean one, so the effective learning rate is unchanged; at 90:1 imbalance a class-balanced sampler would replay the two-image arthropod-pest class roughly 90 times per epoch and overfit it, and weighted loss is the gentler instrument.

4.4 Metrics

Classification is reported as accuracy, macro-F1, weighted-F1, balanced accuracy, and macro precision and recall. Macro averages run over the classes *present in the evaluation split*, with the full declared label space reported alongside: averaging in a zero-support class as a hard 0.0 would describe the split rather than the model, and predictions landing on such classes are still errors and are counted explicitly. Probabilistic quality is reported as negative log-likelihood, Brier score, and expected and maximum calibration error over 15 equal-width bins [14], with the scalar error and the reliability diagram computed from the same bin table so they cannot disagree. Selective prediction [5] is reported as a confidence–abstention curve and the area under the risk–coverage curve. Calibration uses a single temperature fitted by L-BFGS on validation negative log-likelihood and applied unchanged to test and cross-domain logits, never refitted on evaluation data; both variants are reported, which is what makes it possible to ask whether an in-domain fix survives a domain shift.

4.5 Statistical Methodology

Two models evaluated on one dataset are not independent samples, so accuracies are compared with **McNemar’s exact test** on per-item correctness [11,3], in the exact binomial form rather than the chi-square approximation because the discordant counts here are small. Since we report every pairwise comparison,

p -values are also reported with Holm step-down adjustment [9] controlling the family-wise error rate across the whole family of tests.

A paired test computed on one training run is a statement about that run, so every McNemar test is recomputed independently on each seed’s own checkpoints and Holm-adjusted within that seed, and verdicts are reported as holding in all seeds, in none, or in some. We do not pool p -values across seeds: three runs is far too few for a meta-analytic combination, and the question that matters is whether a verdict is stable. Inline p -values are quoted from seed 42; per-seed stability is tabulated in Section 5.

Interval estimates use the **bias-corrected and accelerated (BCa) bootstrap** [4] at the 95% level with 10 000 resamples, computed on seed 42. Where either the bias correction or the acceleration is undefined, as it is for a statistic that does not vary across resamples, the interval degrades to a percentile interval and is labelled as such. The evaluated label space is held fixed at the classes present in the original evaluation, so the statistic keeps one definition across all replicates.

The two instruments answer different questions. The bootstrap interval describes sampling variability of a fixed evaluation set for one checkpoint; the across-seed standard deviation describes variability of the training process itself. A difference smaller than the seed spread is not claimed as a finding regardless of its p -value.

4.6 Reproducibility

Every run revalidates the dataset lock (83449f60136918554904cb9c596122f8b89cb51ff5bb5c52c541394a5698082d) before training begins, so a checkpoint can always be traced to an exact, digest-bound corpus. Runs are sequential on a single Apple-silicon host with a unified-memory accelerator; timings are therefore Apple-silicon numbers and will not transfer to server GPUs, though relative ordering between backbones should. Seeds are set for Python, NumPy, and the framework at the start of every run, and the validation carve-out depends only on the frame, the fraction, and the seed, never on row order or platform. This version reports 18 runs covering seeds 42, 1337, 2026.

A silent mixed-precision failure. Under fp16 autocast the framework skips the optimiser step on non-finite gradients only on CUDA, so on the Metal backend used here one overflowing gradient left a parameter permanently NaN without raising anything. We added the same step skip on the Metal path; the guard leaves a run that never overflows numerically identical, verified by re-training a completed run to a byte-identical checkpoint, so runs before and after remain comparable, and seed 42 was verified unaffected. Runs that skip a step are identified rather than pooled silently: 1 of 10 runs trained under the guard skipped a step, 11 in total out of 38 036: `pv_efficientnet_b0_s1337` (11). Every run was screened for non-finite parameters and for the constant-prediction signature a NaN model produces; none other than that one shows either. The supplement gives the full diagnosis.

5 In-Domain Results

All three backbones saturate the laboratory benchmark. On the PlantVillage test split each reaches accuracy above 0.99 and macro-F1 within a few thousandths of it (Table 1). A $15\times$ span in parameter count buys a difference of a few tenths of a percentage point.

Table 1. In-domain test performance, mean $\pm$ sample standard deviation over three seeds. Weighted-F1, balanced accuracy, and macro precision and recall are in the supplementary table.

Model	Corpus	N	Accuracy	Macro-F1
ResNet-50	PV	10709	0.9945 ± 0.0021	0.9927 ± 0.0029
EfficientNet-B0	PV	10709	0.9964 ± 0.0005	0.9950 ± 0.0006
MobileNetV3-Small	PV	10709	0.9948 ± 0.0004	0.9927 ± 0.0004
ResNet-50	PDC	225	0.6637 ± 0.0403	0.6823 ± 0.0386
EfficientNet-B0	PDC	225	0.6652 ± 0.0128	0.6768 ± 0.0139
MobileNetV3-Small	PDC	225	0.5600 ± 0.0044	0.5876 ± 0.0061

That margin is smaller than the variation the random seed alone produces. Across the 3 seeds the ordering of the three backbones does not reproduce: the backbone that ranks last on two seeds ranks first on the third, and two of the three pairwise comparisons change sign between seeds. Every pairwise verdict on this split is therefore seed-dependent, significant under Holm adjustment on at least one seed and not significant on at least one other, which is the behaviour a benchmark exhibits when the quantity being compared has been exhausted. The supplementary material reports the per-seed orderings rather than a single ranking, which would be an artefact of whichever seed we displayed.

PlantDoc Core, trained and tested in domain, behaves differently, and here the separation is real: MobileNetV3-Small is significantly worse than EfficientNet-B0 on every seed after Holm adjustment and than ResNet-50 on two of the three, with a margin consistent in sign on all three seeds and several times the across-seed spread in both comparisons. The seed that falls short does so on a 225-image split, not by reversing. The two larger backbones remain indistinguishable from each other ($p = 0.7428$, $p_{\text{Holm}} = 1.0000$) and also change sign across seeds. A corpus with headroom separates models the saturated one cannot. Table 2 gives the per-seed stability of every verdict in the paper, and Figure 1 shows all three settings together.

6 Cross-Domain Generalisation

Transferred to field imagery, the same checkpoints lose roughly three quarters of their macro-F1. Relative degradation is $76.9 \pm 1.1\%$ for ResNet-50, $76.9 \pm 0.7\%$ for EfficientNet-B0, and $81.2 \pm 1.1\%$ for MobileNetV3-Small, against a

Table 2. Stability of every pairwise verdict across independent training seeds. Each seed’s McNemar test uses that seed’s own checkpoints and is Holm-adjusted within that seed; ‘sig’ counts the seeds in which the comparison is significant and ‘sgn’ whether the sign of the accuracy difference is the same in all three. Δ is the mean accuracy difference. A verdict holding in every seed is a statement about the architectures; one holding in a single seed is a statement about that run. All three laboratory comparisons are seed-dependent; both verdicts that reproduce everywhere place MobileNet V3-Small last. Per-seed p -values are in the supplementary material.

Evaluation	RN50 vs EB0			RN50 vs MNv3-S			EB0 vs MNv3-S		
	sig	sgn	Δ	sig	sgn	Δ	sig	sgn	Δ
PlantVillage (lab)	1/3	no	-0.0019	2/3	no	-0.0003	2/3	yes	+0.0016
PlantDoc Core (field)	0/3	no	-0.0015	2/3	yes	+0.1037	3/3	yes	+0.1052
PV $\rightarrow$ PDC (shift)	0/3	no	-0.0070	3/3	yes	+0.0345	3/3	yes	+0.0415

Fig. 1. Macro-F1 by backbone and evaluation setting: mean with a bar of one across-seed standard deviation, and each of the 3 seeds drawn beside it. In the laboratory panel one backbone’s spread across seeds covers the distance between backbones, so no ordering read from it is a property of the architecture; the same checkpoints separate cleanly under the shift and on the field test split.

leakage audit that establishes 0 exact overlaps between the two corpora. This is a generalisation measurement, not a contamination artefact. Absolute drops are in the supplementary table.

Absolute cross-domain performance is poor for every backbone (Table 3). The bootstrap intervals on the 1 951-image evaluation set are narrow relative to the collapse itself: on seed 42 the macro-F1 point estimates are 0.2414, 0.2359, and 0.1982, with intervals [0.2253, 0.2595], [0.2203, 0.2538], and [0.1831, 0.2158]. The magnitude is therefore not an artefact of a small evaluation set. For scale, a predictor that always emits the most frequent shared class reaches **[PENDING]** accuracy and **[PENDING]** macro-F1 on the same images, so the models are informative but not by a wide margin.

6.1 What the Ranking Does and Does Not Support

Two of the three pairwise comparisons are stable across every seed. Both larger backbones beat MobileNetV3-Small under the shift, significantly on each seed

Table 3. Cross-domain generalisation on the frozen shared-class PlantDoc Core subset, mean $\pm$ sample standard deviation over three seeds. Metrics cover only the 21 shared classes; unmapped images are excluded rather than counted as errors. ‘Ret. mass’ is the mean source-model probability retained inside the shared space before renormalisation.

Model	N	Accuracy	Macro-F1	Ret. mass
ResNet-50	1951	0.2583 ± 0.0206	0.2291 ± 0.0107	0.6367 ± 0.0945
EfficientNet-B0	1951	0.2653 ± 0.0102	0.2300 ± 0.0065	0.6595 ± 0.0086
MobileNetV3-Small	1951	0.2238 ± 0.0162	0.1869 ± 0.0109	0.5662 ± 0.0477

after family-wise correction ($p = 0.0002$ and 0.0008 on seed 42, $p_{\text{Holm}} = 0.0013$ and 0.0045), with margins several times the across-seed standard deviation.

The comparison between the two larger backbones supports nothing. It is not significant on any seed ($p = 0.7488$, $p_{\text{Holm}} = 1.0000$, on 351 discordant items), the sign of the difference changes between seeds, and the mean margin is an order of magnitude below the seed spread. We therefore decline to order them, and note that a paper reporting a single seed could have ordered them either way with equal apparent justification.

Our ordering claim is therefore a *partial* one: MobileNetV3-Small below both larger backbones. The full three-way ordering reproduces in no setting, cross-domain included, because it inherits the instability of the pair we decline to order, which is why the supplementary ranking table marks every setting unstable.

Set against the in-domain picture, the asymmetry is the result. In domain, where the benchmark is saturated, no ordering survives reseeding, not even the partial one. Under the shift the partial ordering reproduces on every seed. The laboratory benchmark is not merely a weak predictor of field ranking; it does not produce a stable ranking to predict from.

6.2 How the Models Fail

The failure is not refusal but misallocation. Every shared class receives predictions from every backbone, and none is left unpredicted; instead the models over-assign a small number of frequent laboratory classes, with the most heavily over-predicted class absorbing several times its true share of the evaluation set. Mean probability mass retained inside the shared space before renormalisation is 0.637 ± 0.094 , 0.659 ± 0.009 , and 0.566 ± 0.048 : a third or more of each model’s belief falls on classes the protocol never asked about, which is itself a symptom of the shift rather than of the mapping.

7 Calibration and Selective Prediction

In domain, temperature scaling behaves exactly as advertised. The fitted temperature is remarkably consistent across backbones and seeds, 0.7028 ± 0.0076 , 0.7006 ± 0.0040 , and 0.6976 ± 0.0046 , and it reduces expected calibration error by an order of magnitude (Table 4).

Table 4. Expected calibration error for the PlantVillage-trained models, mean $\pm$ sample standard deviation over three seeds. ‘(T)’ applies the temperature fitted on the held-out validation split, unchanged and never refitted on evaluation data; it does not move the argmax, so accuracy is identical with and without it. In domain the standard fix works and ECE falls by an order of magnitude. Carried across the shift it reverses, in all nine model–seed cases. Fitted temperatures are given in the text. PlantDoc-trained calibration, negative log-likelihood, Brier score, maximum calibration error, and AUC are in the supplementary table.

Model	PlantVillage test		PlantDoc Core (shift)	
	uncalibrated	with T	uncalibrated	with T
ResNet-50	0.0987 ± 0.0113	0.0092 ± 0.0011	0.2829 ± 0.0459	0.4178 ± 0.0533
EfficientNet-B0	0.1015 ± 0.0069	0.0095 ± 0.0013	0.2561 ± 0.0186	0.3973 ± 0.0192
MobileNetV3-S	0.0988 ± 0.0025	0.0091 ± 0.0003	0.1517 ± 0.0139	0.2938 ± 0.0128

Carried across the domain shift, the same scalar makes calibration worse. For every backbone and every seed, nine cases without exception, cross-domain ECE rises rather than falls. This is not a marginal degradation: the calibration error after applying the standard remedy is 1.48, 1.55, and 1.94 times what it was without any calibration at all. It is also the most uniform result in the paper, nine model–seed cases moving the same direction, from a fitted temperature whose across-seed standard deviation never exceeds 0.0076.

Selective prediction confirms that the confidence signal degrades rather than merely shifting: the area under the risk–coverage curve rises from 0.0007 ± 0.0002 in domain to 0.6428 ± 0.0282 cross-domain for ResNet-50, the other backbones behaving similarly. Abstention buys far less accuracy out of domain than in it, because the confidences it depends on have stopped being informative.

8 Discussion

8.1 A Saturated Benchmark Carries Little Ranking Information

On seed 42 all three laboratory comparisons are significant, which is why a single-seed reading of these same runs would have reported a clean separation. That separation is an artefact of which seed is displayed, as Figure 1 shows directly.

Of the nine pairwise verdicts in Table 2, three hold on every seed, two fail on every seed, and four depend on the seed. All three laboratory comparisons fall in the seed-dependent group. The fourth is MobileNetV3-Small against ResNet-50 on the field test split, and it is seed-dependent for a different reason: the margin is consistent in sign on every seed and more than three times the across-seed spread, but 225 test images leave one seed short of the Holm threshold. Every verdict that reproduces on all seeds places MobileNetV3-Small last, while ResNet-50 and EfficientNet-B0 are indistinguishable in both field settings with the sign changing between seeds. The result is therefore not that the in-domain

ordering inverts under the shift, but that there is no stable in-domain ordering for the shift to invert.

The practical reading is uncomfortable for a literature organised around in-domain leaderboards. Choosing a backbone by laboratory accuracy, the only signal available without a cross-domain evaluation, ranks MobileNet V3-Small above ResNet-50 on two of the 3 seeds, when under the shift it is significantly worse than ResNet-50 on all three. By the time accuracies exceed 0.99 the remaining differences reflect fit to corpus-specific regularities at least as much as disease recognition [20,6].

8.2 In-Domain Calibration Does Not Transport

Temperature scaling [7] is close to free and close to universal advice: fit one scalar on held-out validation data, apply it, obtain better-calibrated confidences without touching accuracy. In domain it works here exactly as advertised. Across the domain shift it does not merely fail to help, it actively hurts.

The mechanism is legible. Label smoothing leaves the models mildly *under*-confident in domain, so the fitted temperature falls below 1 and *sharpens* the predictive distribution. Sharpening is precisely right in domain, where the model is usually correct. Out of domain the model is already over-confident relative to its much lower accuracy, and sharpening amplifies the error it was supposed to correct. That mechanism also bounds the claim: what we demonstrate is that the standard pipeline as commonly assembled, label smoothing followed by in-domain temperature fitting, yields a scalar that is harmful under shift. Whether a model trained without label smoothing would fit a temperature above 1 and degrade more gracefully is a separate question this experiment does not settle (Section 9).

The deployment consequence inverts the usual recommendation: a temperature fitted on in-domain validation data should not be transported across a domain shift, and transporting it is worse than shipping the model uncalibrated. This is consistent with the broader finding that uncertainty quantification degrades under shift [15], but sharper: the standard fix does not merely lose effectiveness, it changes sign. Because temperature does not move the argmax, a practitioner monitoring only accuracy would never see it happen.

Both findings depend on the controls. Without the leakage audit the honest form of the cross-domain result would be “degradation of at least this much, assuming no contamination”; with 0 exact overlaps established by exhaustive search and human adjudication of every near candidate, the assumption becomes a measurement. And because the shared space comes from a deterministic normaliser and a pre-existing policy rather than from us, its conservative bias makes the headline gap *larger* than a generous mapping would.

9 Limitations and Threats to Validity

1. **The two label spaces differ, and the mapping is conservative.** In-domain evaluation is 38-way, cross-domain 21-way over a restricted space,

so the measured drop combines domain shift with a change in task difficulty and should be read as a paired trend across models rather than as an isolated quantity: the trend is the claim, the absolute magnitude is not. Excluded near-misses are listed with reasons; a human-authored correspondence would likely enlarge the shared space and move the numbers, most plausibly upward, so ours are a lower bound on transfer under a mapping that refuses to guess.

2. **The calibration result is scoped to a label-smoothed pipeline.** Every run uses label smoothing 0.1, which is why the fitted temperature falls below 1 and sharpens rather than softens. The finding is therefore about the standard pipeline as commonly assembled rather than about temperature scaling in isolation; an ablation without label smoothing would separate the two and is the obvious next experiment.
3. **The field corpus carries documented label noise.** Images whose filenames contradict their assigned class are known to exist in PlantDoc. This bounds achievable accuracy on it and is not corrected here, because correcting it would alter the locked corpus and make the evaluation unreproducible against the published lock.
4. **The field test split is small, and so is the design.** With 225 images across 27 evaluable classes, per-class metrics have wide intervals, which is why we report bootstrap intervals, avoid per-class claims, and why one otherwise stable verdict falls short of significance on one seed. Three seeds and three backbones likewise make the ranking-instability finding a demonstration that in-domain rank *can* fail to transfer under a controlled protocol, not an estimate of how often it does. Bootstrap intervals use seed 42 alone and no test incorporates training-process variability, reported separately as across-seed spread, to which we attach no significance claim. Timing figures come from an Apple-silicon accelerator and do not transfer to server GPUs in absolute terms.
5. **Cross-domain evaluation includes field-corpus training images.** All 1951 shared-class images are used, since a laboratory-trained model has seen none of them (Section 3.3). Scoring only PlantDoc’s own test split would give a much smaller set with wider intervals; the split composition is recorded in every result file so that alternative can be reconstructed.
6. **Governance status.** These results rest on a scientific experiment lock binding digests, counts, taxonomy, and splits. It is not a completed formal dataset-governance freeze, and we do not claim otherwise.

10 Conclusion

Closing the two confounds that usually sit between a cross-corpus comparison and its interpretation changes what can be concluded, not merely how carefully it must be hedged. Two findings survive that scrutiny. First, a near-saturated in-domain benchmark carries almost no information about which backbone transfers, because it does not produce an ordering stable enough to carry: across 3

seeds every pairwise verdict on the laboratory split changes with the seed, and the backbone that ranks last on two seeds ranks first on the third. The cross-domain partial ordering separating the smallest backbone from the other two reproduces on every seed. Selecting an architecture on laboratory accuracy is therefore not a conservative default; it is an active bet on a property the benchmark does not measure. Second, temperature scaling fitted on in-domain validation data degrades out-of-domain calibration below the uncalibrated baseline, because label smoothing drives the fitted temperature to sharpen a distribution already overconfident where it matters, and a practitioner watching accuracy alone cannot observe it.

Neither finding requires a new architecture or a new corpus. Both required an evaluation built so that the alternative explanations could be ruled out rather than assumed away, and both point the same direction: for agricultural decision support, the evaluation protocol deserves at least as much scrutiny as the model. The protocol, dataset lock, leakage audit, mapping provenance, and generation scripts for every table and figure are released with the paper.

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